

LATE CRETACEOUS RADIOLARIANS FROM THE NONI FORMATION, WEST TIMOR, INDONESIA

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ABSTRACT

Late Cretaceous (Cenomanian–Turonian) radiolarian fauna was recovered from a single chert sample of the Noni Formation in the Miomaffo District of West Timor, which is generally viewed as part of the allochthonous ‘Banda Terrane’. This fauna is characterized by the presence of *Cryptamphorella conara*, *Diacantocapsa euganea*, *Dictyomitra formosa*, *Dictyomitra montisserei*, *Guttacapsa biacuta*, *Stichomitra communis*, *Patelula verteroensis*, *Pessagnobrachia fabianii*, *Praeconocaryomma lipmanae* and others. The character of the radiolarian fauna shows a close resemblance to those in South Sulawesi and is very different from age-equivalent radiolarian faunas in the ‘autochthonous’ southern foldbelts of West Timor (Kolbano) and Timor Leste (Viqueque). This report on the Late Cretaceous radiolarians in Timor attempts to identify the fauna, to clarify its age, and to indicate the paleogeographic origin of radiolarian-bearing chert of the Noni Formation.

Keywords: Noni Formation, Late Cretaceous radiolarians, West Timor, Timor Leste

INTRODUCTION

Since Tan (1927) described the radiolarian assemblage in Rotti Island, occurrences of radiolarians in West Timor and Rotti Island have largely been considered in any detail only by the present authors since the 1990s. We have discussed radiolarian assemblages within the age ranges Triassic–Late Jurassic (Harsolumakso et al., 1995; Kolbano area), Middle Triassic (Sashida et al., 1999; Kefamenanu area), Late Triassic (Sashida et al., 1996; Nefokoko area; Martini et al., 2000; Niki-Niki, Soe and Kapan areas), Middle Jurassic (Sashida et al., 1999; Rotti Island) and Early Cretaceous (Munasri and Sashida, 2018; Clove, 1997; Munasri, 1998; Kolbano area). In the eastern part of Timor Island (Timor Leste), Middle Jurassic radiolarians were reported by Keep et al. (2009) and discussed by Haig and Bandini (2013).

The present paper concerns Late Cretaceous radiolarians from chert of the Noni Formation, in West Timor (Figure 1). The radiolarian-bearing cherts of the Noni Formation were considered to be of Late Jurassic or Early Cretaceous age (Haile et al., 1979; Earle, 1983). Previously, Audley Charles et al. (1974) suggested a possible Aptian age for the Palelo (Noni Formation) green cherts, unconformably overlain by Turonian (?) cherty limestones and black cherts. Haile et al. (1979) compared the Noni Formation chert sequences to those in the

Bantimala area of South Sulawesi, which were then only dated as broadly Jurassic–Cretaceous in age. Now we know the cherts sequence exposed in the Paring River, Bantimala area of South Sulawesi, is of Albion to Cenomanian (middle Cretaceous) age (Wakita et al., 1994). Haig and Bandini (2013) referred to the Noni Formation as a correlative of a Middle Jurassic black argillite block in mélange sampled near Viqueque, Timor Leste, based on the radiolarian assemblage comparisons that they made with the Middle Jurassic succession in the Kolbano area of West Timor (Harsolumakso et al., 1995). Disputes on the age and origin of the Noni Formation will be discussed further later.

The primary aims of this paper are: (1) to identify the radiolarian fauna, (2) to clarify its age, and (3) to consider the possible paleogeographic origin of the radiolarian chert in the Noni Formation.

AHH (the second author) conducted a field survey and obtained rock samples along the Noil Noni River, in the Miomaffo region of West Timor. The first author (M) then washed one chert sample, NN02AH, to examine the radiolarian content.

GEOLOGICAL SETTING

The Noni Formation, named after the Noil Noni River was described by Rosidi et al. (1979) as a unit of

deep-sea sedimentary rocks comprising well-bedded mainly greenish radiolarian chert, with cherty limestone and clayey chert. The cherts of the Noni

Formation, typically bedded on a scale of 5–15 cm thickness, is highly deformed and folded with irregular fold axis and broken bedding planes.

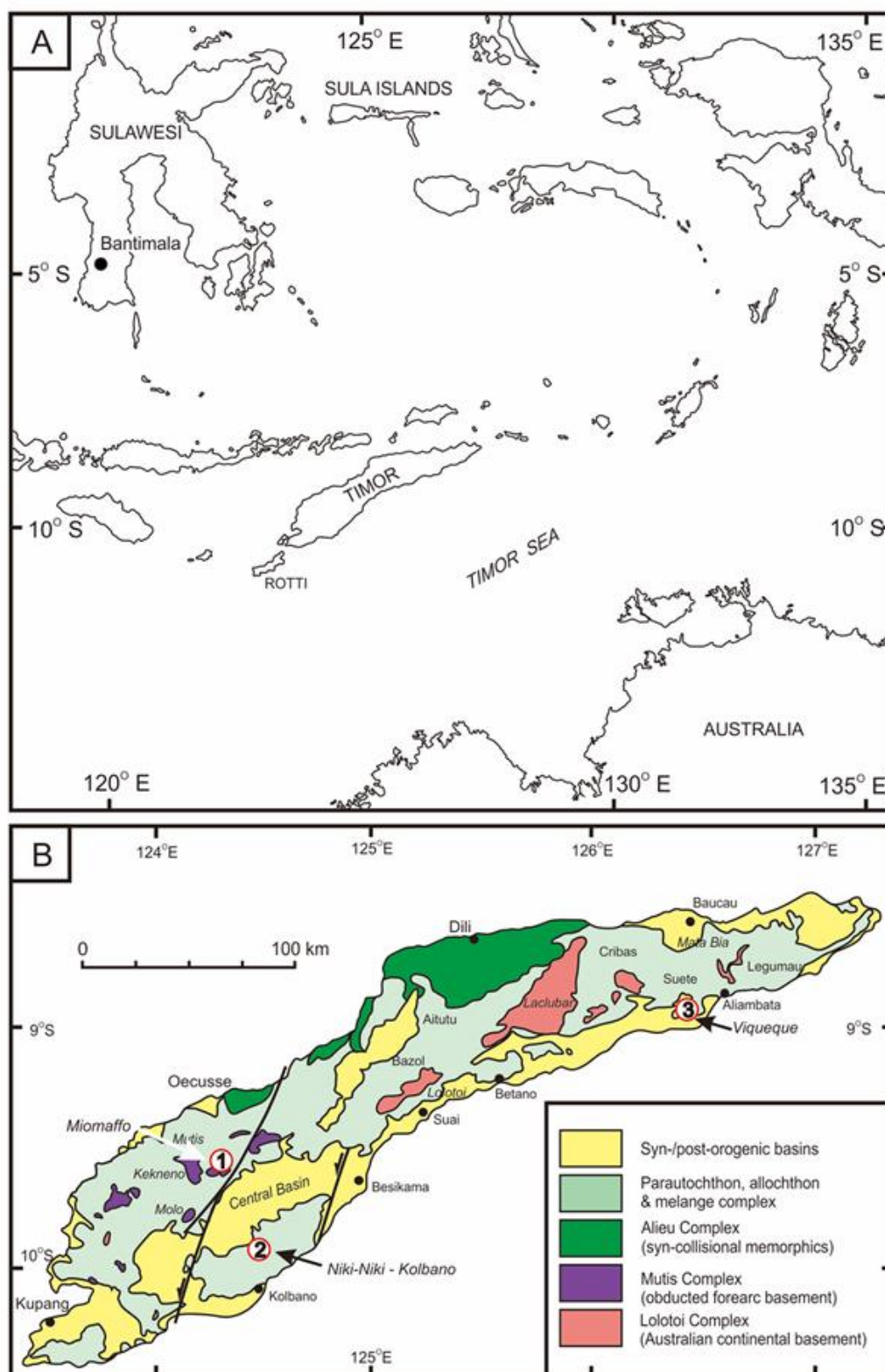


Figure 1: Maps of Eastern Indonesia and Timor Island, showing the location of studies related to radiolaria. (A) Location and names of islands where radiolaria are found as mentioned and compared in this paper: Timor Island (see Figure 1B), Bantimala area, South Sulawesi studied by Wakita et al. (1994), and Sula Islands described by Pessagno and Hull (2002). (B) Locations of radiolarians findings on Timor Island discussed in this paper: (1) Late Cretaceous radiolarians of the Noni Formation in the Miomaffo region. (2) Middle to Late Jurassic radiolarians in the Kolbano to Niki-niki area (Harsolumakso et al., 1995) and (3) Middle Jurassic radiolarians near Viqueque, East Timor (Haig and Bandini, 2013).

Rosidi et al. (1979) proposed the Noni and Haulasi Formations as units together forming the Palelo Group (Palelo Series of Tappenbeck 1940, Van West 1941). The Palelo Group is a Cretaceous to Eocene sedimentary and volcanic succession that overlies the Mutis metamorphic rocks, and comprises a lower deep-water unit and an upper shallow water unit. The lower Palelo Group corresponds to the Noni Formation. Fossil *Globotruncana* sp. found in the cherty limestone indicates a Late Cretaceous age (Rosidi et al., 1979). The upper Palelo Group, corresponding to the Haulasi Formation, consists of conglomeratic graywacke, sandstone, tuffaceous shale, bedded marl, and limestone. Middle Paleocene to Middle Eocene foraminifera (*Nummulites* sp., *Alveolina* sp.) were found in limestones (Rosidi et al., 1979). Preliminary age determinations for the presumed Haulasi Formation in the Samé area of Timor Leste were reported by Charlton et al. (2018). Six genera and one species of dinoflagellates were recorded (*Exochosphaeridium* sp., *Florentinia* sp., *Heterosphaeridium* sp., *Odontochitina* sp., *Palaeohystrichophora infusorioides*, *Spiniferites* sp. and *Subtilisphaera* sp.), all of which have been identified in numerous petroleum exploration wells drilled on the Australian continental margin to the south of Timor. For instance, all seven genera/species occur in the late Albian-early Cenomanian interval in the Mount Ashmore-1B well.

Harris (2006) reported that an andesitic clast in a conglomerate of the typical Haulasi Formation found in the Bebe Susu massif, Timor Leste yielded a U/Pb age of 83 Ma (approximately Santonian-Campanian). The Palelo Group in West Timor is overlain by Eocene deposits of the Metan Formation (Rosidi et al., 1979), which consists of agglomerates with tuffaceous matrix and components of pumice, pyroxene basalt and andesite lava, and andesitic and dacitic tuffs, and represent a part of the Eocene 'Great Indonesian Arc' of Harris (2006). The upper part contains lenses of limestone and sandy marl with Early Eocene foraminifera. A dacitic tuff from the Mosu massif in West Timor yielded an U/Pb age of 35 Ma (approximately Late Eocene) (Harris, 2006).

In terms of tectonostratigraphy, rock assemblages in Timor are frequently described in term of allochthonous, autochthonous and para-autochthonous elements (for details see previous authors e.g. Audley Charles, 1968; Carter et al., 1976; Barber et al., 1977; Charlton et al., 1991; Harsolumakso et al., 1995). The Palelo Group (including the Noni Formation) and the underlying Mutis metamorphic complex (Haile et al., 1979; Earle, 1983) were described as elements of the Banda Terrane (Sopaheluwan, 1990; Harris, 2006) which is interpreted as an allochthonous body overthrust onto the para-autochthonous Australian margin successions of Timor Island.

MATERIAL AND METHODS

One sample of chert, NN02AH originating from the bedded chert layer of the Noni Formation, was collected from the succession exposed in the Noil Noni River near Noil Toko Village, Subdistrict of Miomaffo Barat in the District of Timor Tengah Utara, West Timor (Figure 2 and Figure 3). The chert sample was crushed into 1-2 cm fragments and soaked in dilute 1 % to 5 % hydrofluoric acid (HF) for about 24 hours (Pessagno & Newport, 1972). The acid then was discarded and the sample washed, sieved and dried. Radiolarians within the sample residue were picked under a binocular microscope. Most of the obtained radiolarians have poor to very poor preservation. Radiolarian images presented were photographed using a Scanning Electron Microscope (SEM) in the Geological Department of the Institut Teknologi Bandung, Indonesia. Identified radiolarians are listed in Table 1 and are illustrated in Plates 1 and 2.

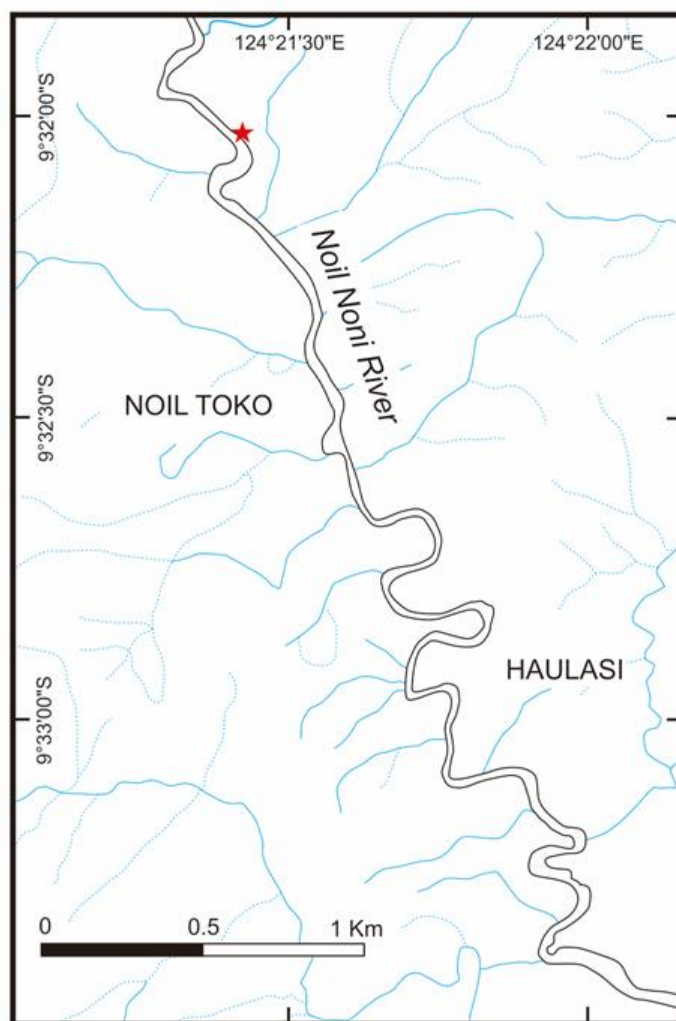


Figure 2: Map showing sample location (red star) for radiolarian-bearing chert in the Noil Noni River in the Miomaffo region, West Timor.



Figure 3: Disharmonic folds in bedded chert of the Noni Formation in the bank of Noil Noni River, West Timor as shown in Figure 2.

RADIOLARIAN FAUNA AND AGE

A total of 35 specimens comprising 15 genera and 1 indeterminate species are presented in this paper. Among them, 11 species are used for age determination: *Cryptamphorella conara* (Foreman), *Diacanthocapsa euganea* Squinabol, *Dictyomitraformosa* Squinabol, *Dictyomitra montisserei* (Squinabol), *Guttacapsa biacuta* (Squinabol), *Squinabollum fossilis* (Squinabol), *Stichomitra communis* Squinabol, *Paronaella communis* (Squinabol), *Pessagnobrachia fabianii* (Squinabol), *Pessagnobrachia irregularis* (Squinabol), and *Praeconocaryomma lipmanae* Pessagno.

From the full faunal list, 11 species were originally described from Italy (Mediterranean region) (Squinabol, 1903, 1904, 1914; O'Dogherty, 1994), and 8 of which can be used for age determination. We use the basic range-chart of O'Dogherty (1994) based on the Unitary Association Zones (U. A. Zones) (Figure 4). As there is no standard radiolarian biochronology available for the Late Cretaceous (Bandini et al., 2006), we sought to compare several other published radiolarian zonation schemes. For this study the biostratigraphic range of certain radiolarian species based on the Unitary Association is compared with those established in the following selected 7 publications: Bak (2011; Italy and Poland), Dumitrica (1970; Romania), Pessagno (1976; California), Sanfilippo & Riedel (1985;

composite), Taketani (1982; Japan), Thurow (1988; Atlantic), and Tumanda (1989; Japan, composite) [Figure 4]. Based on the presence of the above mentioned species, a most likely Cenomanian–Turonian (early Late Cretaceous) age can be assigned for the radiolarian assemblage of the Noni Formation. It is possible that the age is slightly older (Cenomanian until the late Albian), but it will not be younger than Turonian as the last occurrences of *Cryptamphorella conara*, *Diacanthocapsa euganea*, *Stichomitra communis* and *Praeconocaryomma lipmanae* are up to the end of the Turonian (Dumitrica, 1970, Taketani, 1982, Thurow, 1988). Using only the Unitary Association zonation of Dogherty (1994), the age range of the Noni Formation sample is within U.A. Zone 17 (middle Cenomanian). It is possibly one fairly precise age determination within a formation that has an as-yet undetermined broader age range.

Comparing with radiolarian faunas from northern Hokkaido, Japan (Tumanda, 1989), the similar presence of *Stichomitra communis*, *Squinabollum fossilis*, *Dictyomitra densicostata* (here identified as *D. montisserei*), *Stichomitra cf. stocki*, suggests the radiolarian assemblage from the Noni Formation is correlative with her *Thanarla praevenata*–*Holocryptocanium geyserenses* to *Alieveum praegallowayi*–*Amphipyndax* sp. A Assemblage Zones within the Cenomanian to Turonian–Coniacian age range. In Italy 12 species described by Bak (2011), including *Cryptamphorella conara*,

Diacanthocapsa euganea, *Dictomitra formosa*, *Dictyomitra montesserei*, *Guttacapsa biacuta*, *Squinabollum fossilis*, *Stichomitra communis*, *Paronaella communis*, *Patellula verteroensis*, *Pessagnobrachia fabianii*, *Pessagnobrachia irregularis*, and *Praeconocaryomma lipmanae*, of upper Cenomanian–lower Turonian age, are also present in the radiolarian assemblage of the Noni Formation. Using the above studies, we interpret the best fit age for the Noni Formation sample is within the Cenomanian–Turonian (100.5–89.8 Ma; International Commission on Stratigraphy, 2019).

DISCUSSION

The occurrence of well-known species such as *Dictyomitra formosa*, *Diacanthocapsa euganea*, *Stichomitra communis*, *Cryptamphorella conara*, together with *Guttacapsa biacuta* and *Praeconocaryomma lipmanae* undoubtedly determine the Noni Formation as having an early Late Cretaceous age (Cenomanian–Turonian). However, it is not clear yet if this age is representative of the entire chert of the Noni Formation, or whether this chert sequence may have a broader age range.

Haig and Bandini (2013) considered that Middle Jurassic black shales and green to white radiolarian cherts in Harsolumakso et al.'s (1995) Succession E, exposed in the Kolbano region of West Timor are equivalent to the Noni Formation (Palelo Group). However, Harsolumakso et al. (1995) did not interpret their Succession E (dated as Aalenian–Tithonian) as stratigraphic equivalent of the Noni

Formation. Haig and Bandini (2013) furthermore indicated that Succession E of Harsolumakso et al. (1995) has lithological similarity to thin bedded siliceous argillites in large blocks within a mélange unit near Viqueque in Timor Leste. Haig and Bandini (2013) interpreted their Viqueque successions as lithostratigraphically equivalent to Succession E in the Kolbano area (quoted by them as Noni Group). By establishing the Late Cretaceous age for radiolarian cherts of the Noni Formation, equating the Noni Formation with the Succession E of the Kolbano area and correlating it to that in the siliceous argillites mélange block near Viqueque, Timor Leste is no longer appropriate. The siliceous argillite blocks in mélange near Viqueque in Timor Leste may correspond with the black shale of Succession E in the Kolbano area of West Timor (Harsolumakso et al., 1995), but we consider that the Jurassic successions in both Timor Leste (Haig and Bandini, 2013) and Succession E (Harsolumakso et al., 1995) are better assigned to the Wai Luli Formation of Audley-Charles (1968).

The radiolarian fauna collected from the Noni Formation resembles that of the cherts in the Paring River in the Bantimala area of South Sulawesi, interpreted as late Albian–early Cenomanian age by Wakita et al. (1994b: previously examined by Haile et al., 1979). These two assemblages have several distinct taxa in common, such as *Cryptamphorella conara*, *Diacanthocapsa euganea*, *Dictyomitra montisserei* (*Dictyomitra* sp. in Wakita et al., 1994) and *Stichomitra communis*. It seems that deposition of the Noni Formation radiolarian cherts occurred at the same time or soon after deposition of the cherts in Bantimala, South Sulawesi.

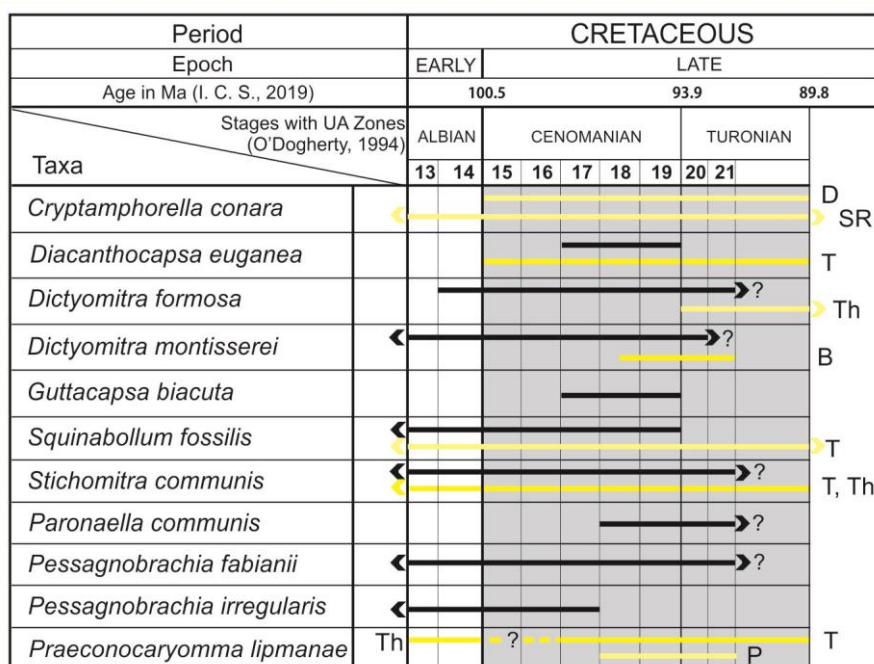


Figure 4: Biostratigraphic range of selected radiolarian species compiled from various authors. Range-chart is adopted from O'Dogherty's (1994) Unitary Association Zones. The age-range indicated by black bars are from O'Dogherty (1994). In yellow are ranges suggested by D: Dumitrica (1970), SR: Sanfilippo and Riedel (1985), T: Taketani (1982), Th: Thurow (1980), B: Bak (2011), P: Pessagno (1976).

Table 1: List of Late Cretaceous radiolarians from chert sample NN02AH of the Noni Formation, West Timor.

Plate 1	Radiolarian species
1.1.	<i>Dictyomitra formosa</i> Squinabol
1.2.	<i>Archaeodictyomitra</i> sp.
1.3.	<i>Stichomitra communis</i> Squinabol
1.4.	<i>Stichomitra communis</i> Squinabol
1.5.	<i>Stichomitra communis</i> Squinabol
1.6.	<i>Archaeodictyomitra</i> sp.
1.7.	<i>Dictyomitra montisserei</i> (Squinabol)
1.8.	<i>Dictyomitra montisserei</i> (Squinabol)
1.9.	<i>Stichomitra stocki</i> (Campbell & Clark)
1.10.	<i>Novixitus</i> sp. cf. <i>N. mclaughlini</i> Pessagno
1.11.	<i>Dictyomitra montisserei</i> (Squinabol)
1.12.	<i>Dictyomitra montisserei</i> (Squinabol)
1.13.	<i>Pseudodictyomitra</i> sp. A.
1.14.	<i>Diacanthocapsa euganea</i> Squinabol
1.15.	<i>Guttacapsa biacuta</i> (Squinabol)
1.16.	<i>Dorypyle elliptica</i> Squinabol
1.17.	<i>Cryptamphorella conara</i> (Foreman)
1.18.	<i>Cryptamphorella</i> sp.
1.19.	<i>Squinabollum fossilis</i> (Squinabol)
Plate 2	Radiolarian species
2.1.	<i>Paronaella communis</i> (Squinabol)
2.2.	<i>Paronaella communis</i> (Squinabol)
2.3.	<i>Pessagnobrachia fabianii</i> (Squinabol)
2.4.	<i>Pessagnobrachia irregularis</i> (Squinabol)
2.5.	<i>Spumellarian</i> gen. et sp. indet.
2.6.	<i>Crucella</i> sp.
2.7.	<i>Crucella</i> sp.
2.8.	<i>Savaryella</i> sp.
2.9.	<i>Triactoma</i> sp. cf. <i>T. cellulosa</i> Foreman
2.10.	<i>Triactoma</i> sp. cf. <i>T. cellulosa</i> Foreman
2.11.	<i>Pseudoacanthosphaera</i> sp. cf. <i>P. spinosissima</i> (Squinabol)
2.12.	<i>Pseudoacanthosphaera</i> sp. cf. <i>P. spinosissima</i> (Squinabol)
2.13.	<i>Pessagnobrachia</i> sp.
2.14.	<i>Patellula verteroensis</i> (Pessagno)
2.15.	<i>Praeconocaryomma lipmanae</i> Pessagno
2.16.	<i>Praeconocaryomma</i> sp. cf. <i>P. universa</i> Pessagno

Munasri and Sashida (2018) reported Early Cretaceous radiolarians from the Nakfunu Formation in the Kolbano area, West Timor. The composition of this fauna is quite different from that found in the Noni Formation (this report) and in the Bantimala area of South Sulawesi (Wakita et al., 1994). Approximately 35% of radiolarian taxa from the Nakfunu Formation have never been reported from the low-paleolatitude belt. Such taxa are attributed as “non-Tethyan fauna” by Baumgartner (1992,

1993). Conversely, radiolarian fauna from the Noni Formation and the Bantimala area of South Sulawesi share similarities with Alpine–Mediterranean, Central Atlantic, Central Pacific and Japanese samples, and are assigned to a “Tethyan fauna.” This suggests the Noni Formation possibly formed at a lower paleolatitude or in a different paleo-oceanic current system from the Nakfunu Formation of the Kolbano area (Munasri and Sashida, 2018).

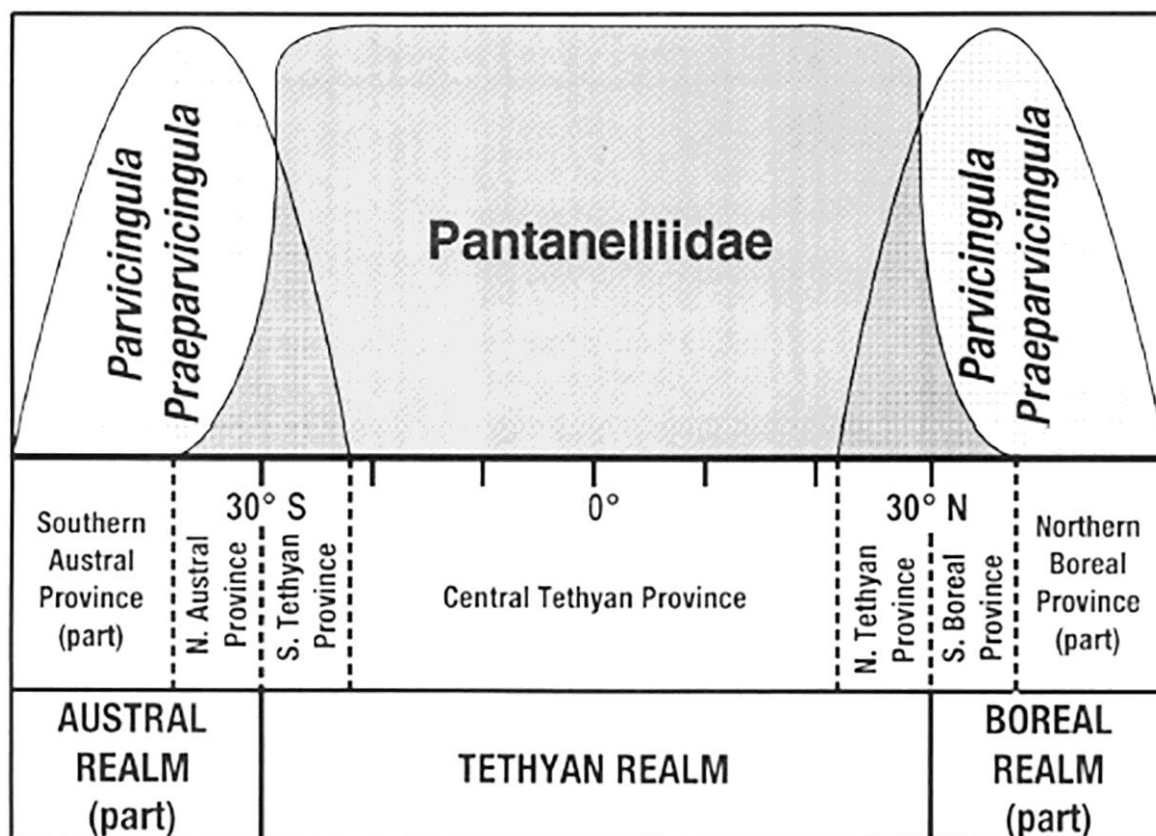


Figure 5: Paleolatitudinal model based on distribution of selected Radiolaria from Jurassic and Lower Cretaceous (adopted from Pessagno and Hull, 2002).

A classification of the paleolatitude zonation of radiolarians, especially for Jurassic faunas, was presented by e.g. Pessagno and Blome (1986), Pessagno and Hull (2002) and Hull (1995). Their paleolatitudinal model describes radiolarian distributions in the Tethyan and Boreal Realms (Figure 5). They suggested that some faunal elements described from Buya Formation assemblages in the Sula Islands are characteristic of the Northern Austral Province (Southern Hemisphere).

Regarding species described from a siliceous argillite block near Viqueque, East Timor (Haig and Bandini, 2013), which contain a common *Parvicingula* / *Praeparvicingula* fauna (one element of which is *Praeparvicingula hurdygurdyensis*), suggests a similarity with that obtained from the Buya Formation in the Sula Islands. In addition, some species in Haig and Bandini (2013)'s assemblage were identical to taxa found by Tan (1927) on Rotti Island, such as *Pseudodictyomitrella* (= *Stichocapsa*) *singularis* Tan and *Pseudodictyomitrella* (= *Cyrtocapsa*) *pseudacerra* Tan. These taxa are not commonly found in the low paleolatitude belt, nor in a Tethyan faunas. We consider that the radiolarian faunas from the siliceous argillite near Viqueque in Timor Leste (Haig and Bandini, 2013) and possibly that from Succession E in the Kolbano area in West Timor (Harsolumakso et al., 1995) derived from the Northern Austral Province (Southern Hemisphere) as defined by Pessagno and Hull (2002). The Middle

Jurassic Circum-Antarctic current conveyed the "non-Tethyan fauna" to the deposition place for the corresponding Wai Luli Formation in the northern Australian shelf and the paleogeographically related para-autochthonous successions of Timor. We consider that the Middle Jurassic radiolarian assemblages mentioned above faced similar occurrences with that fauna obtained from the Nakfunu Formation in West Timor during Early Cretaceous time (Munasri and Sashida, 2018).

Conversely, the Late Cretaceous radiolarian-bearing cherts from the Noni Formation (Paleo Group) were deposited in the low paleolatitude Tethyan realm as part of the Banda Terrane (Harris, 2006) and overthrust as an allochthonous element on Timor (Marks 1961, Carter et al. 1976, Earle 1983).

TAXONOMIC LIST AND REMARKS

The following radiolarian taxa are listed in alphabetical order under Suborders Nassellaria and Spumellaria. Remarks are given where needed, especially to explain the position of taxa, synonym explanation, and description of taxon that has not been previously recognized (e.g. *Pseudodictyomitra* sp. A.).

1. Suborder Nassellaria

Archaeodictyomitra spp.

Plate 1, Figures 2, 6.

Cryptamphorella sp.

Plate 1, Figure 18

Remarks: This form is included in the genus *Cryptamphorella* by its general outline. Test spherical without aperture, showing the opposite side of cephalo-thorax or distal part of inflated abdomen.

Cryptamphorella conara (Foreman, 1968)

Plate 1, Figure 17

Hemicryptocapsa conara Foreman, 1968, p. 35, pl. 4, figs. 11a and 11b

Cryptamphorella conara (Foreman), Dumitrica, 1970, p. 80, pl. 11, figs. 66a–c; Taketani, 1982, p. 67, pl. 7, figs. 6a,b, 7a,b; Sanfilippo and Riedel, 1985, p. 613, text fig. 12.1a–c; Thurow, 1988, p. 399, pl. 1, fig. 2, and pl. 5, fig. 1; Bak, 2011, p. 87, figs. 43B,C.

Remarks: This species is common in the Cenomanian and Turonian (Dumitrica, 1970), spanning from Albian through Maastrichtian (Sanfilippo and Riedel, 1985).

Diacanthocapsa euganea Squinabol, 1903

Plate 1, Figure 14

Diacanthocapsa euganea Squinabol, 1903, p. 133, pl. 8, fig. 26; Taketani, 1982, p. 68, pl. 8, figs. 2a–3b; pl. 12, fig. 15; Tumanda, 1989, p. 36, pl. 7, fig. 5; O'Dogherty, 1994, p. 218, pl. 36. Figs. 19–21; Bak, 2011, p. 94, figs. 45I, J.

Remarks: Three-segmented cryptocephalic nasselarians. Due to poor preservation, our specimen shows an obscure apical horn. This species appears in the middle Cenomanian time (his U.A. Z. 17–19) of the Mediterranean region (O'Dogherty, 1994), Cenomanian – Turonian in Japan (Taketani, 1982).

Dictyomitra formosa Squinabol

Plate 1, Figure 1

Dictyomitra formosa Squinabol, 1904, p. 232, pl. X, fig. 4; Pessagno, 1976, p. 51, pl. 8, figs. 10–12; Taketani, 1982, p. 58, pl. 4, figs. 6a,b and pl. 11, fig. 13; Thurow, 1988, p. 400, pl. 1, fig. 25; Wakita, *et al.*, 1994a, figs. 5.1–3; figs. 6.1–3; O'Dogherty, 1994, p. 80, pl. 4, figs. 8–12; Bak, 2011, p. 98, figs. 47K–M; Asis & Basir Jasin, 2012, p. 90. Pl. 1, fig. 4.

Dictyomitra sp. cf. *D. formosa* Squinabol, Thurow, 1988, p. 400, pl. 1, fig. 23.

Remarks: *D. formosa* Squinabol differs from *D. multicostata* Zittel *sensu stricto* by having deep strictures between the postabdominal chambers, more massive costae, a more markedly lobate test, and only weakly developed costae on the cephalis and thorax.

Dictyomitra montisserei (Squinabol, 1903)

Plate 1, Figures 7–8, 11–12

Stichophormis Montis Serei Squinabol, 1903, p. 137, pl. 8, fig. 38.

Archaeodictyomitra sliteri Pessagno, 1977, p. 43, pl. 6, figs. 3,4, 22, 23, 27; Tumanda, 1989, p. 36, pl. 7, fig. 2.

Dictyomitra sp. A. Taketani, 1982, p. 59, pl. 4, figs. 7a–b.

Dictyomitra densicostata Pessagno, Tumanda, 1989, p. 36, pl. 9, fig. 5.

Dictyomitra montisserei (Squinabol), O'Dogherty, 1994, p. 77, pl. 3, figs. 1–29; Bak, 2011, p. 100, figs. 48A–C.

Remarks: Form of slender, conical to cylindrical multi-segmented nasselarians, with costae throughout. Constrictions weak to well developed. The degree of preservation is very influential in distinguishing *Dictyomitra montisserei*, *Dictyomitra densicostata* and *Archaeodictyomitra sliteri*.

Dorypyle elliptica Squinabol, 1903b

Plate 1, Figure 16

Lithapium ellipticum Squinabol, p. 117, pl. 10, fig. 27.

Dorypyle elliptica Squinabol, O'Dogherty, 1994, p. 206, pl. 33, figs. 8–15.

Guttacapsa biacuta (Squinabol)

Plate 1, Figure 15

Cenellipsis biacutus Squinabol, p. 116, pl. 8, fig. 24.

Guttacapsa biacuta (Squinabol), O'Dogherty, p. 226, pl. 37, figs. 31–35; Bak, 2011, p. 101, fig. 49A.

Pseudodictyomitra sp. A.

Plate 1, Figure 13

Remarks: The only one specimen of multi-segmented *Pseudodictyomitra* with seven post-abdominal chambers. Test elongate, slender, conical. Cephalis smooth, sharply pointed apically. Abdomen faintly costate. Single rings of primary pores are situated in strictures at joints separating adjacent segments; the first one separating thorax and abdomen. The post-abdominal chambers slowly increasing in width distally. Post-abdominal chambers with six to seven elongated thin and weakly costae visible in lateral view. Wide grooves separate the adjacent costae. This form differs from other species of the genus *Pseudodictyomitra* by having wide grooves separating the adjacent costae, and thin and weak costae.

Squinabollum fossilis (Squinabol, 1903)

Plate 1, Figure 19

Clistophaena fossilis Squinabol, 1903, p. 130. pl. , fig. 11.

Sethocapsa sp. A. cf. *S. simplex* Thurow, p. 405, pl. 4, fig. 23.

Squinabollum fossilis (Squinabol), Dumitrica, p. 83, pl. 19, figs. 118a–122; Taketani, 1982, p. 70, pl. 6, figs. 10a–b, 11a–b; pl. 13, figs. 10, 11; Tumanda, 1989, p. 40, pl. 7, fig. 15; pl. 8, fig. 17.

Squinabollum fossile (Squinabol), O'Dogherty, 1994, p. 203, pl. 32, figs. 4–10.

Remarks: This single specimen is not well preserved. This species refers to Dumitrica (1970).

Stichomitra communis Squinabol, 1903
Plate 1, Figures 3–5

Stichomitra communis Squinabol, 1903, p. 141, pl. 8, fig. 40; Taketani, 1982, p. 54, pl. 3, fig. 9, and pl. 11, fig. 5; Thurow, 1988, p. 406, pl. 4, fig. 10; Tumanda, 1989, p. 40, pl. 7, fig. 7; O'Dogherty, 1994, p. 144, pl. 17, figs. 6–16; Bak, 2011, p. 118, figs. 54K, L.

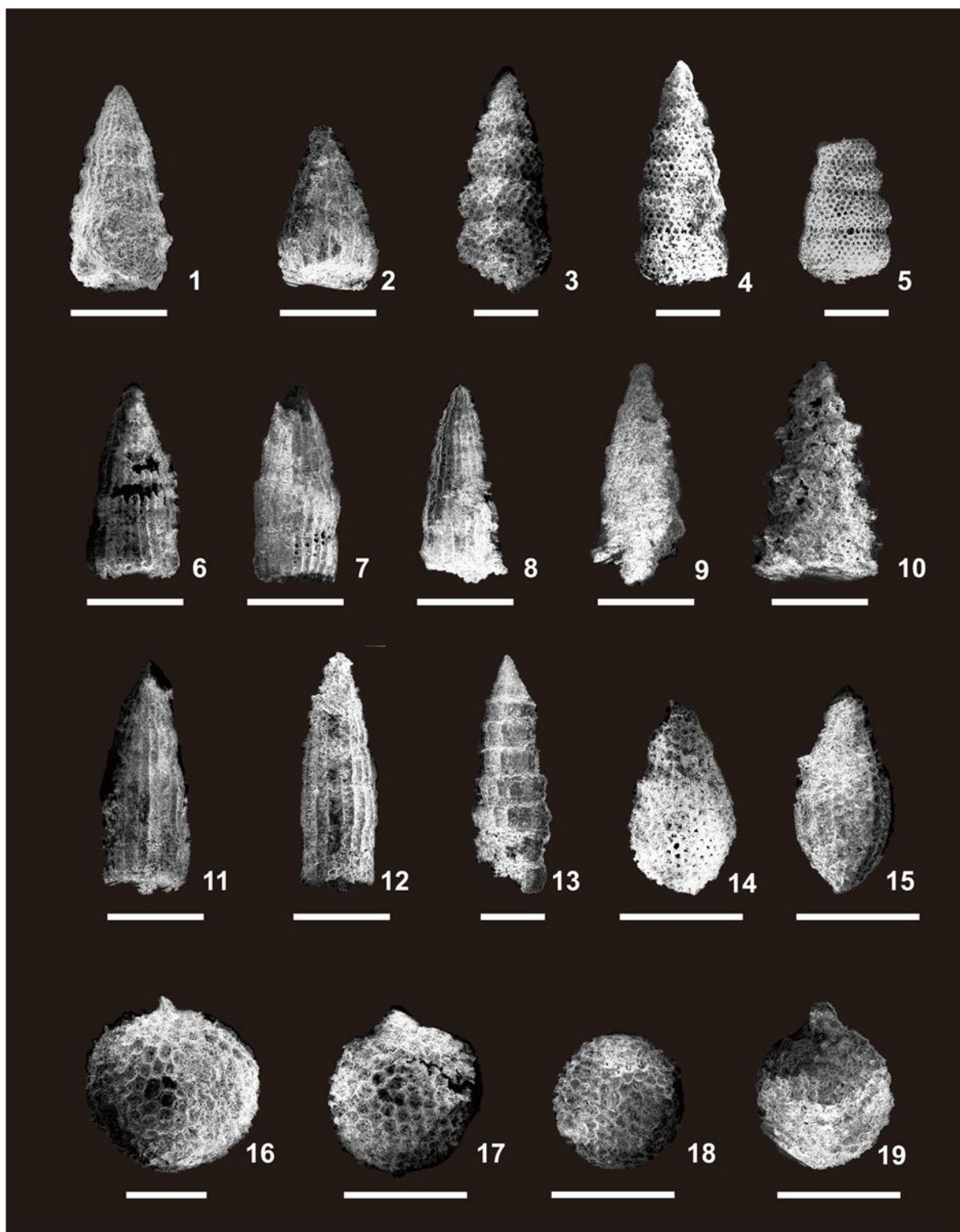


Plate 1. Late Cretaceous radiolarians from the Noni Formation, West Timor. (1) *Dictyomitra formosa*. (2) *Archaeodictyomitra* sp. (3–5) *Stichomitra communis*. (6) *Archaeodictyomitra* sp. (7–8, 11–12) *Dictyomitra montisserei*. (13) *Pseudodictyomitra* sp. A. (14) *Diacanthocapsa euganea*. (15) *Guttacapsa biacuta*. (16) *Dorypyle elliptica*. (17) *Cryptamphorella conara*. (18) *Cryptamphorella* sp. (19) *Squinabollum fossilis*. All images are scanning electron photomicrographs. Bar scales = 100 μ m.

Remarks: This species is recorded from upper Cenomanian to lowermost Coniacian strata of Japan, from Cenomanian to Turonian strata of southern Europe and northwest Africa, and from Albian to Turonian strata at Hole 398D and Oman (Thurrow, 1988); Albian-Cenomanian of Mediterranean region (O'Dogherty, 1994; Umbria-Marche, central Italy (Bak, 2011)).

Stichomitra sp. cf. *S. stocki* (Campbell & Clark, 1944)
Plate 1, Figure 9

Remarks: Poorly preserved specimen. Multi-segmented test, conical to cylindrical distally, with hemispherical cephalis, knob-like. We place this taxon as a form that closely resembles *Stichomitra stocki* by its general outline. *Stichomitra stocki* was re-described by O'Dogherty (1994) as a taxon that many authors subsequently identified as most typical for the genus *Amphipindax* Foreman (1966).

Novixitus sp. cf. *N. mclaughlini* Pessagno, 1977
Plate 1, Figure 10

Remarks: Poorly preserved specimen. This taxon resembles *Novixitus mclaughlini* by its general outline. Test multicystid, elongated, with row of large tubercles, without apical horn.

2. Suborder Spumellaria

Crucella sp.

Plate 2, Figures 6–7

Remarks: Poorly preserved specimens. Four-rayed hagiastrid. Rays equal in length, sharply pointed distally. Central area faintly raised, with central depression. This form is recorded from the Late Triassic to Late Cretaceous.

Paronaella communis (Squinabol, 1903)

Plate 2, Figures 1–2

Spongotropus communis Squinabol, 1903, p. 123, pl. 9, fig. 7.

Paronaella communis (Squinabol, 1903), O'Dogherty, 1994, p. 353, pl. 66, figs. 9–16.

Remarks: Poorly preserved specimens. Three-rayed forms having patagium. Test relatively flattened, and triangular in outline. This form is recorded from late Cenomanian to early Turonian of the Mediterranean region (O'Dogherty, 1994).

Patellula verteroensis (Pessagno, 1963)

Plate 2, Figure 14

Stylospongia verteroensis Pessagno, 1963, p. 199, pl. 3, figs. 1–3; pl. 6, figs. 2–3; pl. 7, figs. 3, 6.

Patellula verteroensis (Pessagno), Thurrow, 1988, p. 403, pl. 2, figs. 19, 20; Tumanda, 1989, p. 34, pl. 9, figs. 15, 16; O'Dogherty, 1994, p. 328, pl. 60, figs. 25, 26; Bak, 2011, p. 146, fig. 67I.

Remarks: Test plane-convex in cross-section, circular in outline, lacking spines. This form is recorded from Campanian sediments in the Caribbean region, North Atlantic, Bavaria, and

Cyprus (Thurrow, 1988), and from the Turonian of the Mediterranean region (O'Dogherty, 1994).

Pessagnobrachia sp.

Plate 2, Figure 13

Remarks: This form of an incomplete specimen, is a distal part of three-rayed *Pessagnobrachia*. Tentatively, this taxon is included in the genus *Pessagnobrachia* (synonym of *Patulibrachium*) following remarks in O'Dogherty (1994).

Pessagnobrachia fabianii (Squinabol, 1914)

Plate 2, Figure 3

Rhopalastrum Fabianii Squinabol, 1914, p. 274, pl. 21, fig. 4.

Pessagnobrachia fabianii (Squinabol), O'Dogherty, 1994, p. 359, pl. 67, figs. 17–25.

Remarks: Poorly preserved specimens. Three-rayed hagiastrid. Rays relatively equal in length, and circular in cross-section. Occurs in the late Albian to early Turonian in the Mediterranean region (O'Dogherty, 1994).

Pessagnobrachia irregularis (Squinabol, 1903)

Plate 2, Figure 4

Rhopalastrum irregulare Squinabol, 1903, p. 122, pl. 9, fig. 10.

Patulibrachium inaequalum Pessagno, 1976, p. 30, pl. 1, fig. 2.

Pessagnobrachia irregularis (Squinabol), O'Dogherty, 1994, p. 360, pl. 68, figs. 1–8.

Remarks: Form of not well preserved specimen. Three-rayed hagiastrid, with unequal interrational angles between rays. Rays very long, always unequal in length. Occurs in the middle Albian to middle Cenomanian of the Mediterranean region (O'Dogherty, 1994).

Praeconocaryomma lipmanae Pessagno (1976)

Plate 2, Figure 15

Praeconocaryomma lipmanae Pessagno, 1976, p. 41, pl. 4, figs. 12, 13; Taketani, 1982, p. 47, pl. 9, fig. 3; Bak, 2011, p. 132, figs. 60B, C.

Conocaryomma lipmanae (Pessagno), Thurrow, 1988, p. 398, pl. 5, Fig. 9.

Remarks: Middle Cenomanian to Turonian of Japan, upper Cenomanian to lower Turonian of California, and upper Albian of the North Atlantic (Thurrow, 1988).

Praeconocaryomma sp. cf. *P. universa* Pessagno (1976)

Plate 2, Figure 16

Remarks: Specimen is not well preserved. Cortical shell with latticed nodes possessing pore frames of variable size, close resemblance in outline to *Praeconocaryomma universa* Pessagno.

Pseudoacanthosphaera sp. cf. *P. spinosissima* (Squinabol, 1904)

Plate 2, Figures 11–12

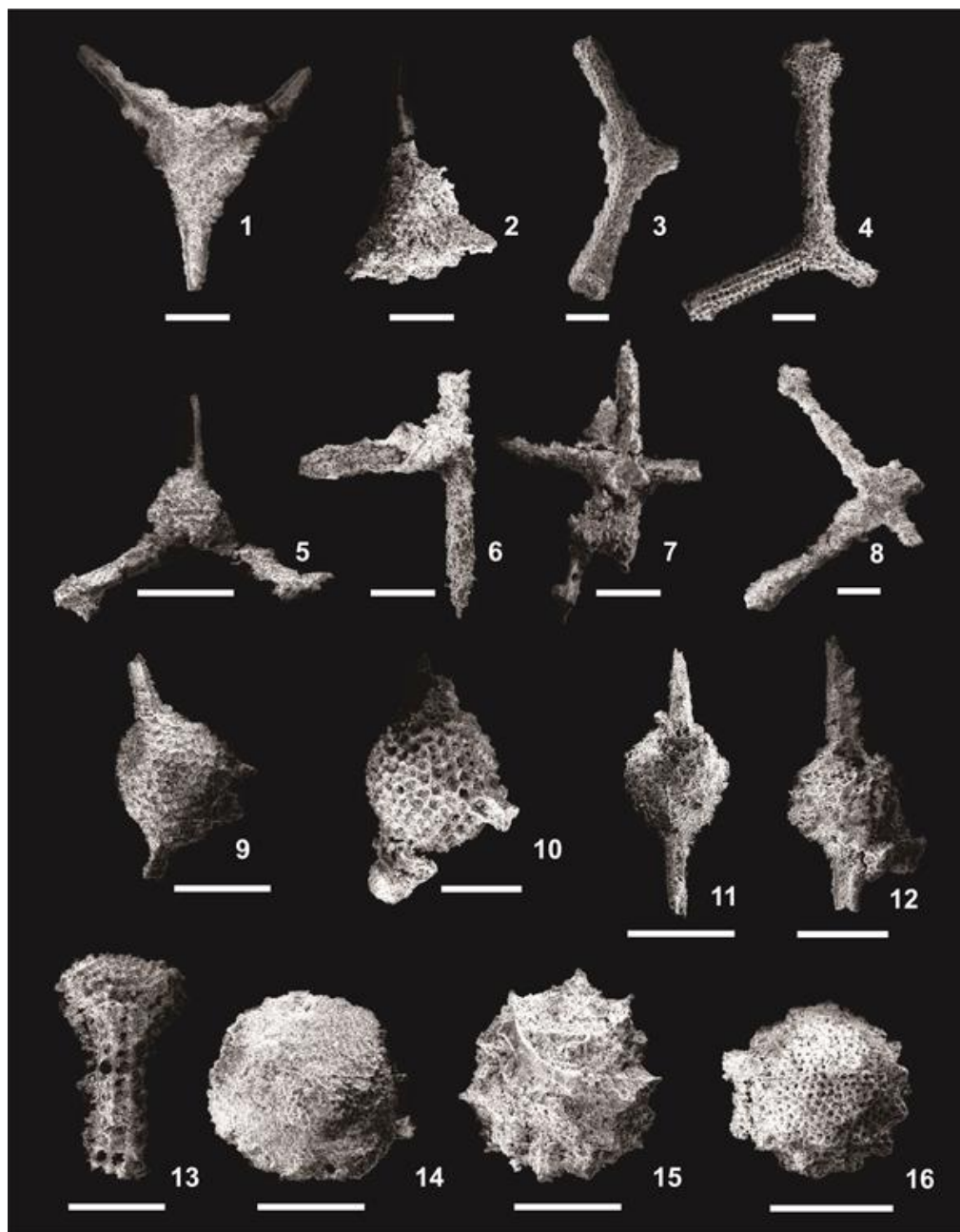


Plate 2. Late Cretaceous radiolarians from the Noni Formation, West Timor. (1–2) *Paronaella communis*. (3) *Pessagnobrachia fabianii*. (4) *Pessagnobrachia irregularis*. (5) *Spumellarian* gen. et. sp. indet. (6–7) *Crucella* sp. (8) *Savaryella* sp. (9–10) *Triactoma* sp. cf. *T. cellulosa*. (11–12) *Pseudoacanthosphaera* sp. cf. *P. spinosissima*. (13) *Pessagnobrachia* sp. (14) *Patellula verteroensis*. (15) *Praeconocaryomma lipmanae*. (16) *Praeconocaryomma* sp. cf. *P. universa*. All images are scanning electron photomicrographs. Bar scales = 100µm.

Remarks: Test with small spinose cortical shell and two moderately long primary spines. Primary spines with three-bladed ridges.

Savaryella sp.
Plate 2, Figure 8

Remarks: Poor preserved specimen of Hagiastrid with four spongy rays. Massive rays with rounded ray tips. This taxon is tentatively included in the genus *Savaryella* Jud (1994). Berriasian to Turonian (O'Dogherty, 1994).

Triactoma sp. cf. *T. cellulosa* Foreman
Plate 2, Figures 9–10

Remarks: Test having subspherical cortical shell, slightly triangular in outline with three bladed primary spines. Three primary spines commonly disposed at 120 degrees. Due to poor preservation, three of the primary spines are broken. Middle

Albian to early Turonian in the Mediterranean region.

Spumellarian gen. et. sp. indet.
Plate 2, Figure 5

Remarks: Spongy cortical shell armed with three primary spines. Due to poor preservation, it seems a slender tiny spine is supported by a pair of sturdy primary spines.

CONCLUSIONS

1. A radiolarian fauna of the Noni Formation was extracted from chert exposed along the Noil Noni River, Miomaffo, West Timor. The fauna is characterized by *Cryptamphorella conara*, *Diacantocapsa euganea*, *Dictyomitra formosa*, *Dictyomitra montisserei*, *Guttacapsa biacuta*,

- Stichomitra communis*, *Patelula verteroensis*, *Pessagnobrachia fabianii*, *Praeconocaryomma lipmanae* and other taxa. One taxon that has not been recognized previously is *Pseudodictyomitra* sp. A.
- The radiolarian assemblage is assigned to the early Late Cretaceous (Cenomanian–Turonian, ~100.5–89.8 Ma according to the geologic timescale of the International Commission on Stratigraphy, 2019).
 - The radiolarian fauna from the Noni Formation has similar characteristic to those found in chert from the Bantimala area (Wakita et al., 1994) of South Sulawesi, both derived from a low paleolatitudinal belt. They differ from those of the Nakfunu Formation in the Kolbano area (Munasri and Sashida, 2018), the black shales and green to white radiolarian chert of Succession E of Harsolumakso et al., 1995) in South West Timor, and the thin-bedded siliceous argillites found as large blocks in mélange near Viqueque, Timor Leste (Haig and Bandini, 2013), all of which have a larger influence of “non-Tethyan fauna” and were derived from the Northern Austral Province (Southern Hemisphere) as defined by Pessagno and Hull (2002).
 - The Late Cretaceous radiolarian-bearing chert of the Noni Formation is interpreted to be deposited in the Tethyan realm or low paleolatitudinal belt, as a part of the ‘allochthonous’ Banda Terrane.
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